



MAX-M Navigation Module



MAX-M, consisting of a navigation GNSS module and an external connection module, ensures reliable geolocation across various fields of application, offering an optimal balance of functionality and cost.

Radio specifications

Operating bands:

- GPS: L1C/A, L1PY, L2C, L2P, L5
- GLONASS: L1CA, L2CA, L2P, L3 CDMA
- BeiDou: B1I, B1C, B2a, B2b, B2I, B3
- Galileo: E1, E5a, E5b, E5 AltBoc, E6
- QZSS: L1C/A, L1 C/B, L2C, L5
- NavIC: L5
- SBAS: EGNOS, WAAS, GAGAN, MSAS, SDCM (L1, L5)
- L-band

Number of processing channels: 448

Number of antenna inputs: 2

Maximum external antenna supply current:
150 mA

Power supply

Input supply voltage: 8 to 48 V

Maximum module power consumption: 1.6 W

Data transmission protocols

- Septentrio Binary Format (SBF)
- NMEA 0183 v2.3, v3.03, v4.0
- RINEX v2.x, v3.x
- RTCM v2.x, v3.x (including MSM)
- CMR v2.0 (output/input), CMR+ (input only)

Interfaces

- 2 UART, TTL 3V3
- USB Type-C

Additional outputs

EVENT 3V3, PPS 3V3

Bit rate

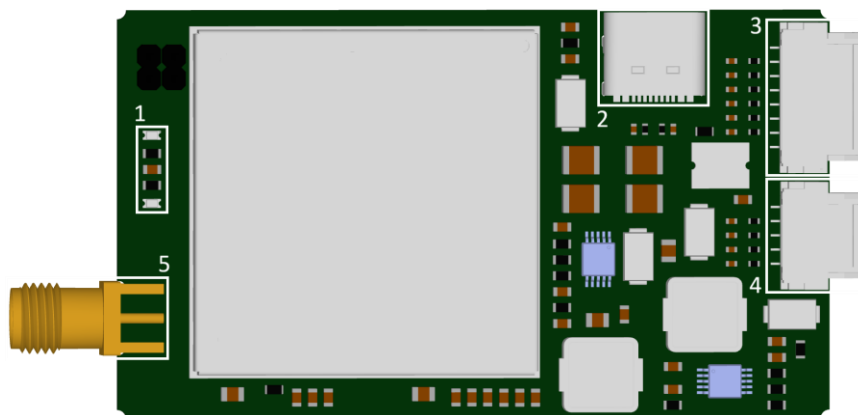
- Factory setting: 115,200 baud
- Configurable range: 1,200–4,000,000 baud

Physical and operational parameters

Operating temperature range: –40 °C to +85 °C

Storage temperature range: –55 °C to +85 °C

External connection module components

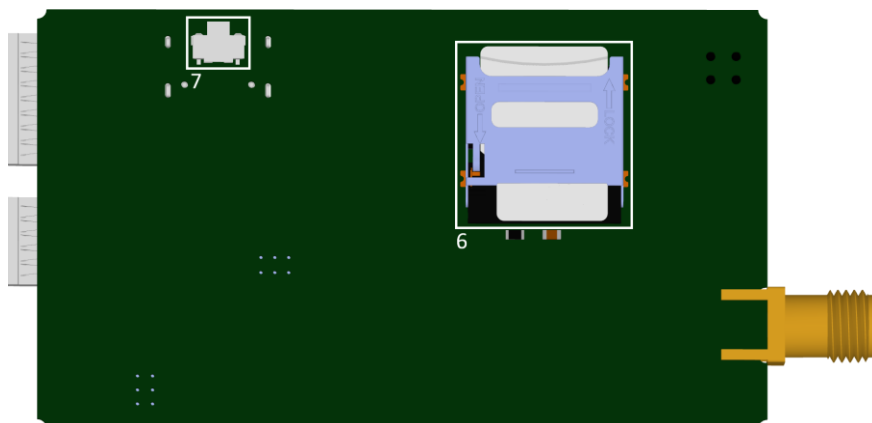


- 1—Indicator zone
- 2—USB Type-C connector
- 3—JST connector, GH series, 1.25 mm pitch, 8 pins
- 4—JST connector, GH series, 1.25 mm pitch, 5 pins
- 5—MMCX connector



MAX-M Navigation Module

Additional components (upon request to the manufacturer)



6—microSD* slot
7—Log button

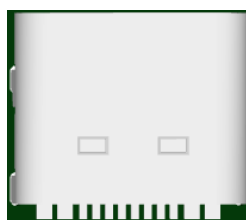
* Supported microSD
cards: 128 GB.

1. Indicator zone



- [Y]—PWR
- [G]—RDY

2. USB Type-C connector



Pinout

- 1—A4, B9—Vbus red
2—B4, A9—Vbus red
3—A6—D+ White
4—B6—D+ White 5—
A7—D- Green 6—
B7—D- Green 7—A1,
B12—GND black
8—B1, A12—GND black

3. JST connector, GH series,
1.25 mm pitch, 8 pins



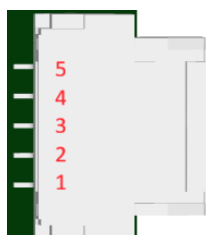
Pinout

- 8—GND
7—PPS
6—EVENT
5—RTS_1
4—CTS_1
3—RxD_1
2—TxD_1
1—+8÷48V

Wire colors

- 8—Black
7—White
6—Brown
5—Blue
4—Yellow
3—Purple
2—Green
1—Red

4. JST connector, GH series,
1.25 mm pitch, 5 pins



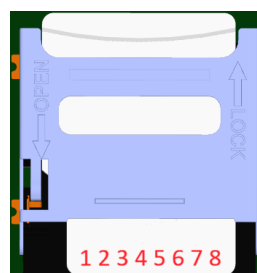
Pinout

- 5—GND
4—RTS_2
3—CTS_2
2—RxD_2
1—TxD_2

Wire colors

- 5—Black
4—Blue
3—Yellow
2—Purple
1—Green

5. microSD slot



Pinout

- 1—Dat2
2—CD
3—Cmd
4—Vdd
5—Clk
6—Vss
7—Dat0
8—Dat1